

Final Assessment-Alternative Assessment

Semester	202520	Division	CIS
Assessment title in Syllabus	Final_Project_Portfolio	Program	Bachelor of Information Technology/ Bachelor of Information Systems

Course Code	CDS 2413		
Course Title	INTRODUCTION TO DATA SCIENCE		
Course Instructor		CRN	
Assessment	Project Deliverable 1 – Data Analysis Report	Submission Date	Week 6

Student Declaration:

Academic Integrity Statement

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

إفادة النزاهة الأكاديمية

وفقاً لسياسة كليات التقنية العليا للنزاهة الأكاديمية

- على الطلبة الالتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبين وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض إلى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my/our own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.



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CLO1: INTRODUCTION AND DATA WRANGLING

1.1 PURPOSE OF DATA ANALYSIS

The purpose of this data analysis is to understand the factors that affect student performance and identify meaningful patterns in the dataset. By analyzing variables such as study hours, attendance, and subject scores, we aim to determine how these factors influence overall academic performance.

This analysis helps:

- Identify key factors affecting student results
- Understand relationships between variables
- Prepare the dataset for machine learning

Example:

Students who study more hours and have higher attendance are expected to achieve higher overall scores.

1.2 TYPE OF PROGRAMMING

This project uses **Python programming** for data analysis.

Python is chosen because:

- It is simple and easy to use
- it is widely used in data science
- It supports powerful libraries

Libraries used:

- **Pandas** - for handling and cleaning data
- **NumPy** - for numerical calculations
- **Matplotlib / Seaborn** - for visualization
- **Scikit-learn** - for machine learning

Python is also used in course lectures and lab exercises, making it the most suitable choice.



NumPy



seaborn

matplotlib

1.3 TYPE AND PURPOSE OF ML ALGORITHM

The machine learning technique selected is **Regression (Supervised Learning)**.

Regression is used when the output variable is numerical. In this project, it will be used to predict student performance.

Purpose:

- Predict overall student performance
- Understand how different variables affect results

Example:

Predicting overall score based on study hours and attendance percentage.

1.4 IDENTIFY THE VARIABLES

Variables are the features used in the dataset.

1. Dependent Variable

- overall_score

This is the main output variable that represents student performance. It is numeric, which makes it suitable for descriptive statistics, correlation analysis, regression, and hypothesis testing.

2. Independent Variables

- study_hours
- attendance_percentage
- math_score
- science_score
- english_score
- age

3. Justification

These variables are selected because they directly influence student performance and help explain variations in overall scores. Academic factors such as subject scores and study habits are strong indicators of student success.

1.5 SAMPLING

Sampling is used to select a smaller portion of the dataset for analysis. It helps reduce processing time while still maintaining accuracy.

1. Random Sampling

Random sampling selects data randomly.

Justification:

- Reduces bias
- Ensures every record has equal chance

2. Systematic Sampling

Systematic sampling selects records using a fixed interval.

Justification:

- Simple and efficient
- Ensures evenly distributed data

CLO2: DESCRIPTIVE, EXPLORATORY AND STATISTICAL ANALYSIS

2.1 PURPOSE OF DESCRIPTIVE ANALYSIS

Descriptive analysis is the first step in understanding the dataset. It helps summarize data, identify patterns, and detect outliers before applying machine learning models.

In this project it helps:

- Understand distribution of student performance
- Identify average and extreme values
- Prepare data for further analysis

Example:

Finding the average overall score and identifying students who perform above or below average.

2.2 SCRIPT FOR DESCRIPTIVE ANALYSIS

A Python function is developed to calculate descriptive statistics for a selected column.

Purpose:

- Calculates detailed statistics
- Reusable for any variable
- Improves efficiency

```
import pandas as pd

df = pd.read_csv("student_performance.csv")

def descriptive_stats(data, column):
    series = data[column].dropna()

    stats = {
        "Mean": series.mean(),
        "Median": series.median(),
        "Mode": series.mode().tolist(),
        "Min": series.min(),
        "Max": series.max(),
        "Range": series.max() - series.min(),
        "Standard Deviation": series.std(),
        "Variance": series.var(),
        "Q1": series.quantile(0.25),
        "Q2 (Median)": series.quantile(0.50),
        "Q3": series.quantile(0.75),
        "IQR": series.quantile(0.75) - series.quantile(0.25),
        "Skewness": series.skew()
    }

    return pd.Series(stats)

result = descriptive_stats(df, "overall_score")
print(result)
```

```
*** Mean          64.006172
    Median         64.2
    Mode          [100.0]
    Min           14.5
    Max           100.0
    Range          85.5
    Standard Deviation 18.932025
    Variance       358.421553
    Q1             49.0
    Q2 (Median)     64.2
    Q3             79.0
    IQR            30.0
    Skewness       -0.044512
    dtype: object
```

2.3 RANDOM SAMPLING

A random sample of 150 records is created and analyzed.

```
sample_150 = df.sample(n=150, random_state=42)
sample_150.head()
```

	student_id	age	gender	school_type	parent_education	study_hours	attendance_percentage	internet_access	travel_time	extra_activities	study_method
6868	6869	18	other	private	phd	6.3	82.9	yes	30-60 min	no	notes
24016	12456	15	other	public	high school	2.1	76.1	yes	<15 min	no	group study
9668	9669	15	male	public	graduate	3.7	84.1	no	<15 min	no	textbook
13640	13641	16	male	public	no formal	2.4	62.3	yes	30-60 min	yes	notes
14018	14019	17	other	private	high school	1.0	91.0	yes	>60 min	yes	online

```
random_stats = descriptive_stats(sample_150, "overall_score")
print(random_stats)
```

	overall_score
Mean	63.274
Median	65.1
Mode	[32.6, 46.3, 56.0, 65.3, 65.7, 65.8, 66.4, 74.0]
Min	18.2
Max	100.0
Range	81.8
Standard Deviation	20.130899
Variance	405.253078
Q1	47.325
Q2 (Median)	65.1
Q3	80.45
IQR	33.125
Skewness	-0.069721
dtype:	object

2.4 SYSTEMATIC SAMPLING

A systematic sample is created using a fixed interval.

```
step = len(df) // 150
systematic_sample = df.iloc[::step].head(150)
systematic_sample.head()
```

	student_id	age	gender	school_type	parent_education	study_hours	attendance_percentage	internet_access	travel_time	extra_activities	study_method
0	1	14	male	public	post graduate	3.1	84.3	yes	<15 min	yes	notes
166	167	15	other	public	post graduate	0.7	63.4	yes	30-60 min	no	group study
332	333	15	other	public	post graduate	3.5	63.8	no	<15 min	no	textbook
498	499	19	female	private	phd	5.4	87.6	yes	30-60 min	yes	notes
664	665	17	female	public	no formal	3.5	77.5	yes	30-60 min	no	mixed

Next steps: [Generate code with systematic_sample](#) [New interactive sheet](#)

```
systematic_stats = descriptive_stats(systematic_sample, "overall_score")
print(systematic_stats)
```

	overall_score
Mean	65.119333
Median	65.0
Mode	[100.0]
Min	27.8
Max	100.0
Range	72.2
Standard Deviation	18.907116
Variance	357.47902
Q1	50.75
Q2 (Median)	65.0
Q3	81.05
IQR	30.3
Skewness	-0.052469
dtype:	object

2.5 SYSTEMATIC SAMPLING

The descriptive statistics of **overall_score** provide important insights into student performance.

- The **mean** represents the average student performance
- The **standard deviation** shows how much scores vary
- The **quartiles** divide students into performance groups
- The **IQR** shows the spread of middle values
- The **skewness** indicates distribution shape

Interpretation:

- If skewness > 0 - **more students scored below average**
- If skewness < 0 - **more students scored above average**

This analysis helps understand the overall academic performance trend.

2.6 VISUALISATION THROUGH VARIOUS CHARTS

Visualization helps identify patterns and relationships in the dataset.

Boxplot:

```
import matplotlib.pyplot as plt
import seaborn as sns

# Style
sns.set(style="whitegrid", palette="pastel")

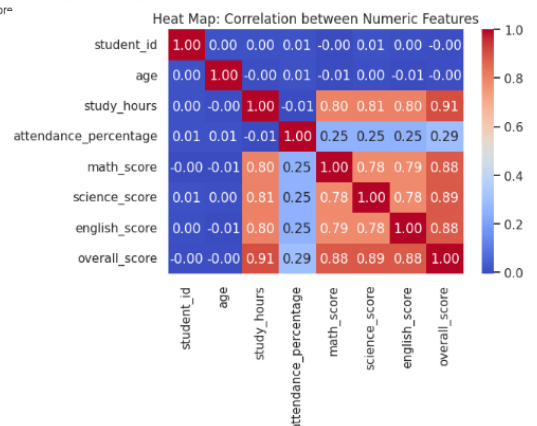
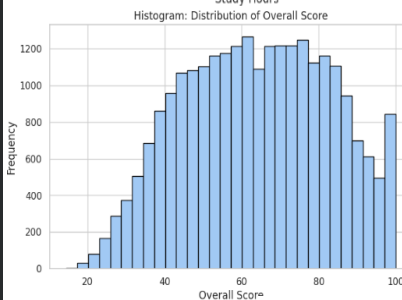
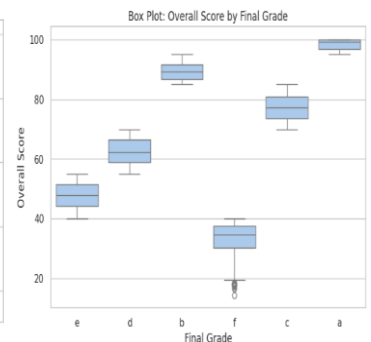
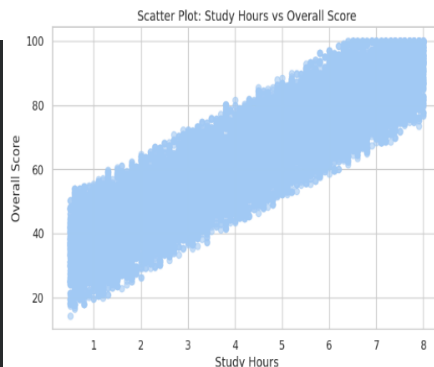
# ----- Scatter Plot -----
plt.figure(figsize=(8,5))
plt.scatter(df["study_hours"], df["overall_score"], alpha=0.6)
plt.title("Scatter Plot: Study Hours vs Overall Score")
plt.xlabel("Study Hours")
plt.ylabel("Overall Score")
plt.show()

# ----- Box Plot -----
plt.figure(figsize=(8,5))
sns.boxplot(x="final_grade", y="overall_score", data=df)
plt.title("Box Plot: Overall Score by Final Grade")
plt.xlabel("Final Grade")
plt.ylabel("Overall Score")
plt.show()

# ----- Histogram -----
plt.figure(figsize=(8,5))
plt.hist(df["overall_score"], bins=30, edgecolor='black')
plt.title("Histogram: Distribution of Overall Score")
plt.xlabel("Overall Score")
plt.ylabel("Frequency")
plt.show()

# ----- Heat Map -----
plt.figure(figsize=(6,4))
numeric_df = df.select_dtypes(include=["int64", "float64"])
corr_matrix = numeric_df.corr()

sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f")
plt.title("Heat Map: Correlation between Numeric Features")
plt.show()
```



2.7 HYPOTHESIS TEST

To analyze the relationship between variables, hypothesis testing is performed using Pearson, Spearman, and Chi-square tests.

1. Pearson Correlation (Numerical vs Numerical):

We test the relationship between **study_hours** and **overall_score**.

```
[16] ✓ 0s  from scipy.stats import pearsonr

r, p = pearsonr(df["study_hours"], df["overall_score"])

print("Pearson Correlation:", r)
print("p-value:", p)

if p < 0.05:
    print("Significant relationship between Study Hours and Overall Score")
else:
    print("No significant relationship between Study Hours and Overall Score")
```

▼ ... Pearson Correlation: 0.9057714454088278
p-value: 0.0
Significant relationship between Study Hours and Overall Score

Interpretation:

The Pearson correlation test is used to measure the relationship between study hours and overall score.

- If the p-value is less than **0.05**, the null hypothesis is **rejected**
- This indicates a significant relationship

This shows whether study hours affect student performance.

2. Spearman Correlation (Ordinal vs Numerical)

We test the relationship between **parent_education** and **overall_score**.

```
[17]: from scipy.stats import spearmanr\n\neducation_map = {\n    "none": 0,\n    "high school": 1,\n    "diploma": 2,\n    "bachelor": 3,\n    "master": 4,\n    "phd": 5\n}\n\ndf["parent_education_num"] = df["parent_education"].str.lower().map(education_map)\n\ndata = df[["parent_education_num", "overall_score"]].dropna()\n\nrho, p = spearmanr(data["parent_education_num"], data["overall_score"])\n\nprint("Spearman Correlation:", rho)\nprint("p-value:", p)\n\nif p < 0.05:\n    print("Significant relationship between Parent Education and Overall Score")\nelse:\n    print("No significant relationship between Parent Education and Overall Score")\n\n... Spearman Correlation: 0.003965077183550175\np-value: 0.656318978133548\nNo significant relationship between Parent Education and Overall Score
```

Interpretation:

The Spearman test is used because parent education is an ordinal variable.

- If **p-value** < 0.05 - significant relationship
- If **p-value** ≥ 0.05 - no relationship

This checks if higher parent education leads to better student performance.

3. Chi-Square Test (Categorical vs Categorical)

We test the relationship between **final_grade** and **gender**.

```
[18]: from scipy.stats import chi2_contingency\n\ntable = pd.crosstab(df["final_grade"], df["gender"])\n\nchi2, p, dof, expected = chi2_contingency(table)\n\nprint("Chi-square:", chi2)\nprint("p-value:", p)\n\nif p < 0.05:\n    print("Significant relationship between Final Grade and Gender")\nelse:\n    print("No significant relationship between Final Grade and Gender")\n\n... Chi-square: 18.14632631270717\np-value: 0.052544920828117625\nNo significant relationship between Final Grade and Gender
```

Interpretation:

The Chi-square test is used to check association between two categorical variables.

- If **p-value** < 0.05 - variables are related
- If **p-value** ≥ 0.05 - no relationship

This shows if gender affects final grade.

2.8 ONE SAMPLE T-TEST

The one-sample t-test is used to check whether the sample represents the population.

```
[19] ✓ Os from scipy.stats import ttest_1samp

population_mean = df["overall_score"].mean()

t_stat, p = ttest_1samp(sample_150["overall_score"], population_mean)

print("Population Mean:", population_mean)
print("Sample Mean:", sample_150["overall_score"].mean())
print("T-statistic:", t_stat)
print("p-value:", p)

if p > 0.05:
    print("Sample represents the population")
else:
    print("Sample does NOT represent the population")

... Population Mean: 64.006172
Sample Mean: 63.274
T-statistic: -0.4454465349254425
p-value: 0.6566436056014051
Sample represents the population
```

Interpretation:

The one-sample t-test compares the sample mean with the population mean.

- If **p-value** > 0.05 - sample represents population
- If **p-value** < 0.05 - sample is significantly different

This confirms whether the random sample is reliable.

END OF DOCUMENTATION.